

CSCI E-102 Econometrics and Causal Inference with R
Part I of Assignment 9

Assume that data are generated as follows:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n,$$

where u_i is the error term such that $\text{Cov}(X_i, u_i) \neq 0$, i.e. X_i is *endogenous*.

Let Z_i be an *instrumental* variable (IV), that is, assume that the following two conditions hold:

1. *Instrument relevance*: $\text{Cov}(Z_i, X_i) \neq 0$.
2. *Instrument exogeneity*: $\text{Cov}(Z_i, u_i) = 0$.

Prove that

$$\beta_1 = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)}.$$

The image shows a handwritten derivation of the IV estimator formula on a grid background. At the top, the formula $\beta_1 = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)}$ is written. Below it, an arrow points to the equation $\text{Cov}(Z_i, Y_i) = \text{Cov}(Z_i, \beta_0 + \beta_1 X_i + u_i) = 0 + \beta_1 \text{Cov}(Z_i, X_i) + 0$. Annotations include a curved arrow labeled "correlated" between Z_i and X_i , and another curved arrow labeled "not correlated" between Z_i and u_i . Below this, the equation $\frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)} = \beta_1 \frac{\text{Cov}(Z_i, X_i)}{\text{Cov}(Z_i, X_i)}$ is written. Finally, the formula $\beta_1 = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)}$ is circled.

$$\beta_1 = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)}$$
$$\rightarrow \text{Cov}(Z_i, Y_i) = \text{Cov}(Z_i, \beta_0 + \beta_1 X_i + u_i) = 0 + \beta_1 \text{Cov}(Z_i, X_i) + 0$$
$$\frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)} = \beta_1 \frac{\text{Cov}(Z_i, X_i)}{\text{Cov}(Z_i, X_i)}$$
$$\beta_1 = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)}$$